MTH207 2022S2 Exam

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* MTH207 Linear Algebra
* MTH219 Fundamentals of Statistics and Probability
* MTH220 Statistical Methods and Inference

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SINGAPORE UNIVERSITY OF SOCIAL SCIENCES

MTH207

Timed Online Assignment - July Semester 2022

Linear Algebra

Tuesday, 15 November 2022 **04:00 pm - 06:00 pm**

Time allowed: 2 hours

INSTRUCTIONS TO STUDENTS:

1. This Timed Online Assignment (TOA) contains **FIVE (5)** questions and comprises **EIGHT (8)** pages (including cover page).

2. You must answer **ALL** questions.

3. If you have any queries about a question, or believe there is an error in the question, briefly explain your understanding and assumptions about that question before attempting it.

4. You MUST **submit your answers via Canvas (similar to TMA submission) at the end time of this TOA** (as stated on this cover page). The 15 minutes grace period as shown on Canvas is strictly meant for technical issues encountered during submission. Thereafter, you will not be able to submit your answers and you will be considered as having withdrawn from the course. **No appeal will be allowed.**

5. You are to include the following particulars in your submission: TOA Course Code, your Full Name, and SUSS PI Number. Name your submission file as TOACourseCode_FullName_StudentPI. Use underscore and Not Space. *Example: TOAXYZ_RaphaelLee_T1923161 (omit D/O, S/O).*

6. Your submission should consist of only one file and **must not exceed 500MB** in size. The file must be a PDF or Microsoft Word file saved in .docx format. All answers are to be typed. Flowcharts and graphs may be scanned or photographed and embedded in the PDF or Word file provided it does not exceed the file size limit of 500MB.

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7. For answers which cannot be typed and require to be hand-written, ensure that question number is clearly stated on each page. All uploaded hand-written answers must be clear, readable and complete. Marks will not be awarded for un-readable or incomplete images.

8. To prevent plagiarism and collusion, your submission will be reviewed by Turnitin. The Turnitin report will only be made available to the marker and you will not be able to view it.

9. The University takes plagiarism and collusion seriously, and your Turnitin report will be examined thoroughly as part of the marking process.

10. Marks will only be awarded if FULL working is shown.

Answer all questions. (Total 100 marks)

Question 1

Let A be a 3×3 matrix satisfying the following three conditions:

1. A has eigenvalues of -1 and 2.
2. The eigenvalue -1 corresponds to eigenvector $\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$.
3. The eigenvalue 2 corresponds to eigenvectors $\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$.

Question 1a

Determine the matrix A. Show your working.

(5 marks)

Question 1b

Calculate an orthonormal basis of the eigenspace associated to the eigenvalue 2.

(5 marks)

Question 1c

Suppose B is another 3×3 matrix such that

$$\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$$

are also eigenvectors of B. Show that A and B commute.

(5 marks)

Question 1d

Calculate the determinant of matrix A.. Determine if the matrix A found in Question 1(a) is invertible.

(5 marks)

Question 2

Consider the following system of linear equations.

$$\begin{aligned} x_1 - x_2 - x_3 - 3x_4 &= 0 \\ x_1 - 2x_2 + x_3 &= 0 \\ 3x_2 + x_3 + 2x_4 + x_5 &= 0 \\ 4x_1 + x_2 - 6x_4 + 2x_5 &= 0 \end{aligned}$$

Question 2a

Write the system of equations in the form of $Ax = 0$ where A is a matrix, x and 0 are vectors. Compute the reduced row echelon form of A.

(5 marks)

Question 2b

Solve the system of linear equations.

(5 marks)

Question 2c

Compute a basis of the null space of A.

(5 marks)

Question 2d

Determine if the vector

$$\begin{pmatrix} 2 \\ 0 \\ -1 \\ 5 \\ -3 \end{pmatrix}$$

is in the null space of A.

(5 marks)

Question 3

Let

$$B = \begin{pmatrix} 2 & 3 & -1 \\ 1 & -2 & 2 \\ 4 & -1 & 3 \end{pmatrix}.$$

Question 3a

Calculate the eigenvalues of B and eigenvectors corresponding to each eigenvalue of B. Write down a basis for the column space of B.

(10 marks)

Question 3b

Show that every vector of $\mathbb{R}^3$ can be written uniquely as a sum $v_1 + v_2$ where v_1 lies in the null space of B and v_2 lies in the column space of B.

(4 marks)

Question 3c

Define a linear transformation $T: \mathbb{R}^3 \to \mathbb{R}^3$ by $T(v_1 + v_2) = v_2$ where v_1 lies in the null space of B and v_2 lies in the column space of B. Note that T is well-defined by the conclusion in Question 3(b). Compute the matrix representation of T with respect to the standard basis of $\mathbb{R}^3$. Give a geometric interpretation of T.

(6 marks)

Question 4

Let

$$v_1 = \begin{pmatrix} 1 \\ 1 \\ 3 \\ 3 \end{pmatrix}, v_2 = \begin{pmatrix} -1 \\ 1 \\ -1 \\ -3 \end{pmatrix}, v_3 = \begin{pmatrix} 2 \\ 5 \\ 10 \\ 6 \end{pmatrix}, v_4 = \begin{pmatrix} 1 \\ 2 \\ 4 \\ p \end{pmatrix}.$$

Question 4a

Determine whether $v_1,\ v_2,\ \text{and}\ v_3$ are linearly independent. Show your working.

(8 marks)

Question 4b

Determine the value(s) of p such that the set $\{v_1, v_2, v_3, v_4\}$ is linearly independent.

(8 marks)

Question 4c

What is the dimension of the subspace that $v_2\ \text{and}\ v_3$ span?

(4 marks)

Question 5a

See the chessboard below in Figure Q5(a). The horizontal and vertical dimensions of all cells are the same. A ball is located at point S. We want to move the ball to point E step by step.

For each step, there are only 2 options.

- Option 1: Moving across 1 cell along +x direction, 2 cells along +y direction.
- Option 2: Moving across 2 cells along +x direction, 1 cell along +y direction.

These 2 options are illustrated in Figure Q5(a) below, starting from point S. If the ball hits the boundary of the chessboard except point E, it returns to point S.

Can the ball reach point E eventually? If yes, show a path from S to E. If no, justify your answer.

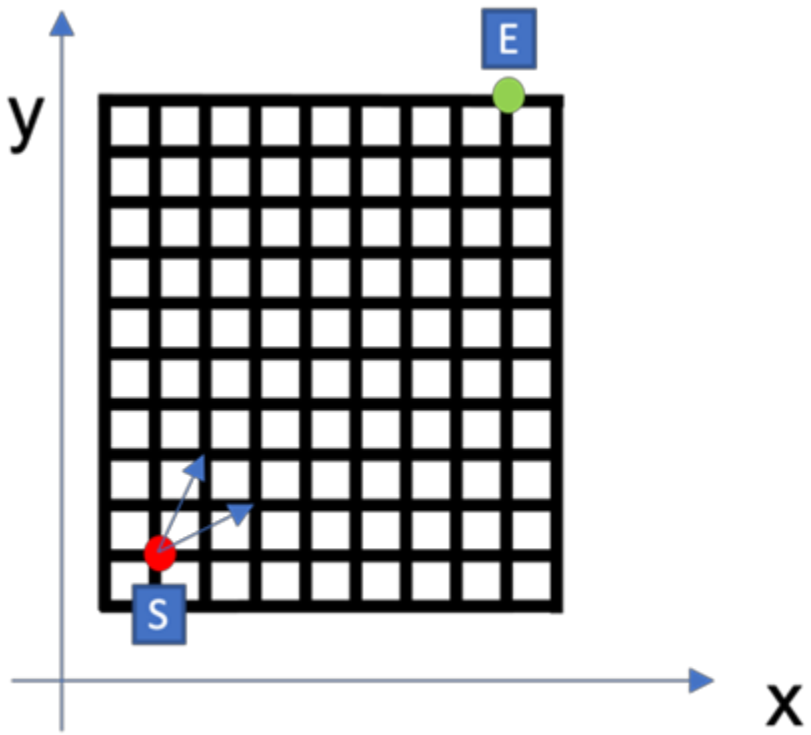

Figure Q5(a)

(10 marks)

Question 5b

Solve the intersection of the following two planes in R^3.

$$\begin{aligned} x - 2y - z &= 0 \\ 3x - 3y + z &= 0 \end{aligned}$$

(10 marks)

----- END OF PAPER -----

1(a) $\lambda_{1,1} = -1 \qquad v_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$

$\lambda_{2,2} = 2 \qquad v_{2,1} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \; v_{2,2} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$

$\Leftrightarrow D = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}, \quad P = \begin{bmatrix} 1 & 1 & 0 \\ 2 & 1 & 1 \\ 3 & 0 & 1 \end{bmatrix}$

To find P^{-1}

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 2 & 1 & 1 & 0 & 1 & 0 \\ 3 & 0 & 1 & 0 & 0 & 1 \end{array}\right] \xrightarrow[r_3 = r_3 - 3r_1]{r_2 = r_2 - 2r_1} \left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & -1 & 1 & -2 & 1 & 0 \\ 0 & -3 & 1 & -3 & 0 & 1 \end{array}\right]$$

$$\xrightarrow[r_3 = r_3 - r_2]{r_2 = -r_2} \left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & -1 & 2 & -1 & 0 \\ 0 & 0 & -2 & 3 & -3 & 1 \end{array}\right]$$

$$\xrightarrow[r_2 = r_2 + \frac{1}{2}r_3]{r_3 = -\frac{1}{2}r_3} \left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1/2 & 1/2 & -1/2 \\ 0 & 0 & 1 & -3/2 & 3/2 & -1/2 \end{array}\right]$$

$$\xrightarrow{r_1 = r_1 - r_2} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1/2 & -1/2 & 1/2 \\ 0 & 1 & 0 & 1/2 & 1/2 & -1/2 \\ 0 & 0 & 1 & -3/2 & 3/2 & -1/2 \end{array}\right] \quad \therefore P^{-1} = \frac{1}{2}\begin{bmatrix} 1 & -1 & 1 \\ 1 & 1 & -1 \\ -3 & 3 & -1 \end{bmatrix}$$

$\therefore A = P D P^{-1}$

$$= \begin{bmatrix} 1 & 1 & 0 \\ 2 & 1 & 1 \\ 3 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} -1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix} \times \frac{1}{2}\begin{bmatrix} 1 & -1 & 1 \\ 1 & 1 & -1 \\ -3 & 3 & -1 \end{bmatrix}$$

$$= \frac{1}{2}\begin{bmatrix} 1 & 3 & -3 \\ -6 & 10 & -6 \\ -9 & 9 & -5 \end{bmatrix}$$

1(b) eigenspace $= \text{span}\{x_1, x_2\}$ where $x_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$, $x_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$

using Gram-Schidmt Process

$$v_1 = x_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \qquad |v_1| = \sqrt{2} \qquad \frac{1}{|v_1|} v_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

$$v_2 = x_2 - \frac{x_2 \cdot v_1}{v_1 \cdot v_1} v_1 = \begin{bmatrix} -1/2 \\ 1/2 \\ 1 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix} \qquad |v_2| = \frac{1}{2}\sqrt{6} = \sqrt{\frac{6}{4}} = \sqrt{\frac{3}{2}}$$

$$\frac{1}{|v_2|} v_2 = \sqrt{\frac{2}{3}} \cdot \frac{1}{2} \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix} = \sqrt{\frac{2}{3} \cdot \frac{1}{4}} \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix} = \frac{1}{\sqrt{6}} \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix}$$

in orthonormal basis $= \left\{ \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \frac{1}{\sqrt{6}} \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix} \right\}$

(c) $B = PDP^{-1}$ where $P = \begin{bmatrix} 1 & 1 & 0 \\ 2 & 1 & 1 \\ 3 & 0 & 1 \end{bmatrix}$, $D = \begin{bmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{bmatrix}$

$$= \frac{1}{2} \begin{bmatrix} a+b & -a+b & a-b \\ 2a+b-3c & -2a+b+3c & 2a-b-c \\ 3a-3c & -3a+3c & 3a-c \end{bmatrix}$$

$$AB = \frac{1}{2} \begin{bmatrix} 1 & 3 & -3 \\ -6 & 10 & -6 \\ -9 & 9 & -5 \end{bmatrix} \times \frac{1}{2} \begin{bmatrix} a+b & -a+b & a-b \\ 2a+b-3c & -2a+b+3c & 2a-b-c \\ 3a-3c & -3a+3c & 3a-c \end{bmatrix}$$

$$= \frac{1}{2} \begin{bmatrix} a+b & -a+b & a-b \\ 2a+b-3c & -2a+b+3c & 2a-b-c \\ 3a-3c & -3a+3c & 3a-c \end{bmatrix}$$

$$BA = \frac{1}{2} \begin{bmatrix} a+b & -a+b & a-b \\ 2a+b-3c & -2a+b+3c & 2a-b-c \\ 3a-3c & -3a+3c & 3a-c \end{bmatrix} \times \frac{1}{2} \begin{bmatrix} 1 & 3 & -3 \\ -6 & 10 & -6 \\ -9 & 9 & -5 \end{bmatrix}$$

$$= \frac{1}{2} \begin{bmatrix} a+b & -a+b & a-b \\ 2a+b-3c & -2a+b+3c & 2a-b-c \\ 3a-3c & -3a+3c & 3a-c \end{bmatrix}$$

since $AB = BA$, then A, B are commutable

For A, B to commute, $A+B = B+A$ and $AB = BA$. Since addition operation is already known to commute, then there is only need to check whether $AB = BA$ since matrix multiplication is generally not commutable.

1(d) $\det A = \det\left(\frac{1}{2}\begin{bmatrix} 1 & 3 & -3 \\ -6 & 10 & -6 \\ -9 & 9 & -5 \end{bmatrix}\right)$

$$= \left(\frac{1}{2}\right)^3 \begin{vmatrix} 1 & 3 & -3 \\ -6 & 10 & -6 \\ -9 & 9 & -3 \end{vmatrix}$$

$$= \frac{1}{8}\left[\begin{vmatrix} 10 & -6 \\ 9 & -5 \end{vmatrix} - 3\begin{vmatrix} -6 & -6 \\ -9 & -5 \end{vmatrix} - 3\begin{vmatrix} -6 & 10 \\ -9 & 9 \end{vmatrix}\right]$$

$$= \frac{1}{8}\left[4 - 3(-24) - 3(36)\right]$$

$$= -\frac{32}{8}$$

$$= -4$$

Since $\det(A) \neq 0$, A is invertible

2(a)
$$\begin{bmatrix} 1 & -1 & -1 & -3 & 0 \\ 1 & -2 & 1 & 0 & 0 \\ 0 & 3 & 1 & 2 & 1 \\ 4 & 1 & 0 & -6 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\left[\begin{array}{ccccc|c} 1 & -1 & -1 & -3 & 0 & 0 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 0 & 3 & 1 & 2 & 1 & 0 \\ 4 & 1 & 0 & -6 & 2 & 0 \end{array}\right] \xrightarrow[r_4 = r_4 - 4r_1]{r_2 = r_2 - r_1} \left[\begin{array}{ccccc|c} 1 & -1 & -1 & -3 & 0 & 0 \\ 0 & -1 & 2 & 3 & 0 & 0 \\ 0 & 3 & 1 & 2 & 1 & 0 \\ 0 & 5 & 4 & 6 & 2 & 0 \end{array}\right]$$

$$\xrightarrow[\substack{r_3 = r_3 + 3r_2 \\ r_4 = r_4 + 5r_2}]{r_2 = -r_2} \left[\begin{array}{ccccc|c} 1 & -1 & -1 & -3 & 0 & 0 \\ 0 & 1 & -2 & -3 & 0 & 0 \\ 0 & 0 & 7 & 11 & 1 & 0 \\ 0 & 0 & 14 & 21 & 2 & 0 \end{array}\right] \xrightarrow[r_4 = 2r_3 - r_4]{r_3 = \frac{1}{7} r_3} \left[\begin{array}{ccccc|c} 1 & -1 & -1 & -3 & 0 & 0 \\ 0 & 1 & -2 & -3 & 0 & 0 \\ 0 & 0 & 1 & 11/7 & 1/7 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \end{array}\right]$$

$$\xrightarrow[\substack{r_2 = r_2 + 3r_4 \\ r_1 = r_1 + 3r_4}]{r_3 = r_3 - \frac{11}{7} r_4} \left[\begin{array}{ccccc|c} 1 & -1 & -1 & 0 & 0 & 0 \\ 0 & 1 & -2 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1/7 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \end{array}\right] \xrightarrow[r_1 = r_1 + r_3 + r_2 + 2r_3]{r_2 = r_2 + 2r_3} \left[\begin{array}{ccccc|c} 1 & 0 & 0 & 0 & 3/7 & 0 \\ 0 & 1 & 0 & 0 & 2/7 & 0 \\ 0 & 0 & 1 & 0 & 1/7 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \end{array}\right]$$

(b)
$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} (-3/7)\, x_5 \\ (-2/7)\, x_5 \\ (-1/7)\, x_5 \\ 0 \\ x_5 \end{bmatrix} = \frac{x_5}{7} \begin{bmatrix} -3 \\ -2 \\ -1 \\ 0 \\ 7 \end{bmatrix}$$

2(c)

$$\text{Basis of } \operatorname{nul} A = \left\{ \begin{bmatrix} -3 \\ -2 \\ -1 \\ 0 \\ 7 \end{bmatrix} \right\}$$

(d)

$$\begin{bmatrix} 2 \\ 0 \\ -1 \\ 5 \\ -3 \end{bmatrix} \neq k \begin{bmatrix} -3 \\ -2 \\ -1 \\ 0 \\ 7 \end{bmatrix} \quad \text{where } k \in \mathbb{R}$$

$\therefore$ The vector isn't in the null space of A.

3(a) let $\det(B - \lambda I) = 0$

$$\begin{vmatrix} 2-\lambda & 3 & -1 \\ 1 & -2-\lambda & 2 \\ 4 & -1 & 3-\lambda \end{vmatrix} = 0$$

$$(2-\lambda)\begin{vmatrix} -2-\lambda & 2 \\ -1 & 3-\lambda \end{vmatrix} - \begin{vmatrix} 3 & -1 \\ -1 & 3-\lambda \end{vmatrix} + 4\begin{vmatrix} 3 & -1 \\ -2-\lambda & 2 \end{vmatrix} = 0$$

$$-\lambda^3 + 3\lambda^2 + \lambda = 0$$
$$-\lambda(\lambda^2 - 3\lambda - 1) = 0$$
$$-\lambda\left(\left(\lambda - \tfrac{3}{2}\right)^2 - \tfrac{13}{4}\right) = 0$$

$$\therefore \lambda_1 = 0, \quad \lambda_2 = \frac{3+\sqrt{13}}{2}, \quad \lambda_3 = \frac{3-\sqrt{13}}{2}$$

case $\lambda = \lambda_1$:

$$\left[\begin{array}{ccc|c} 2-\lambda & 3 & -1 & 0 \\ 1 & -2-\lambda & 2 & 0 \\ 4 & -1 & 3-\lambda & 0 \end{array}\right] \xrightarrow{\lambda = \lambda_1} \left[\begin{array}{ccc|c} 2 & 3 & -1 & 0 \\ 1 & -2 & 2 & 0 \\ 4 & -1 & 3 & 0 \end{array}\right]$$

$$\xrightarrow{\substack{r_1 = \frac{1}{2}r_1 \\ r_2 = r_2 - \frac{1}{2}r_1 \\ r_3 = r_3 - 2r_1}} \left[\begin{array}{ccc|c} 1 & 3/2 & -1/2 & 0 \\ 0 & -7/2 & 5/2 & 0 \\ 0 & -7 & 5 & 0 \end{array}\right]$$

$$\xrightarrow{\substack{r_2 = -\frac{2}{7}r_2 \\ r_3 = r_3 - 2r_2}} \left[\begin{array}{ccc|c} 1 & 3/2 & -1/2 & 0 \\ 0 & 1 & -5/7 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$$\xrightarrow{r_1 = r_1 - \frac{3}{2}r_2} \left[\begin{array}{ccc|c} 1 & 0 & 4/7 & 0 \\ 0 & 1 & -5/7 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

The pivot columns are 1, 2

$\Rightarrow$ c1 and c2 are linearly independant

$$\therefore \text{basis for col } B = \left\{ \begin{bmatrix} 2 \\ 1 \\ 4 \end{bmatrix}, \begin{bmatrix} 3 \\ -2 \\ -1 \end{bmatrix} \right\}$$

3(a) case $\lambda = \lambda_1$:

$$\therefore \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_3 \begin{bmatrix} -4/7 \\ 5/7 \\ 1 \end{bmatrix} = \frac{x_3}{7} \begin{bmatrix} -4 \\ 5 \\ 7 \end{bmatrix}$$

$$\therefore \text{eigenvector } v_1 = \begin{bmatrix} -4 \\ 5 \\ 7 \end{bmatrix}$$

case $\lambda_2 = \frac{3+\sqrt{13}}{2}$:

$$\left[\begin{array}{ccc|c} 2-\lambda & 3 & -1 & 0 \\ 1 & -2-\lambda & 2 & 0 \\ 4 & -1 & 3-\lambda & 0 \end{array}\right] \xrightarrow{\lambda = \lambda_2} \left[\begin{array}{ccc|c} \frac{1}{2}-\frac{\sqrt{13}}{2} & 3 & -1 & 0 \\ 1 & -\frac{7}{2}-\frac{\sqrt{13}}{2} & 2 & 0 \\ 4 & -1 & \frac{3}{2}-\frac{\sqrt{13}}{2} & 0 \end{array}\right]$$

$r_1 = (\frac{1}{2}-\sqrt{13})^{-1} r_1$

$r_2 = r_2 - (\frac{1}{2}-\sqrt{13})^{-1} r_1$

$r_3 = r_3 - 4(\frac{1}{2}\sqrt{3})^{-1} r_1$

$$\left[\begin{array}{ccc|c} 1 & -\frac{\sqrt{13}}{2}-\frac{1}{2} & \frac{1}{6}+\frac{\sqrt{13}}{6} & 0 \\ 0 & -3 & \frac{11}{6}-\frac{\sqrt{13}}{6} & 0 \\ 0 & 1+2\sqrt{13} & \frac{5}{6}-\frac{7\sqrt{13}}{6} & 0 \end{array}\right]$$

$r_2 = -\frac{1}{3} r_2$

$r_3 = r_3 + \frac{1+2\sqrt{13}}{3} r_2$

$$\left[\begin{array}{ccc|c} 1 & -\frac{\sqrt{13}}{2}-\frac{1}{2} & \frac{1}{6}+\frac{\sqrt{13}}{6} & 0 \\ 0 & 1 & -\frac{11}{18}+\frac{\sqrt{13}}{18} & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$r_1 = r_1 - (-\frac{\sqrt{13}}{2}-\frac{1}{2}) r_2$

$$\left[\begin{array}{ccc|c} 1 & 0 & \frac{2}{9}-\frac{\sqrt{13}}{9} & 0 \\ 0 & 1 & -\frac{11}{18}+\frac{\sqrt{13}}{18} & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$$\therefore \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_3 \begin{bmatrix} \frac{\sqrt{13}}{9}-\frac{2}{9} \\ \frac{11}{18}-\frac{\sqrt{13}}{18} \\ 1 \end{bmatrix} = \frac{x_3}{18} \begin{bmatrix} 2\sqrt{13}-4 \\ 11-\sqrt{13} \\ 18 \end{bmatrix}$$

$$\therefore \text{eigenvector } v_2 = \begin{bmatrix} 2\sqrt{13}-4 \\ 11-\sqrt{13} \\ 18 \end{bmatrix}$$

3(a) case $\lambda = \lambda_3$:

$$\left[\begin{array}{ccc|c} 2-\lambda & 3 & -1 & 0 \\ 1 & -2-\lambda & 2 & 0 \\ 4 & -1 & 3-\lambda & 0 \end{array}\right] \xrightarrow{\lambda = \lambda_3} \left[\begin{array}{ccc|c} \frac{1}{2}+\frac{\sqrt{13}}{2} & 3 & -1 & 0 \\ 1 & -\frac{7}{2}+\frac{\sqrt{13}}{2} & 2 & 0 \\ 4 & -1 & \frac{3}{2}+\frac{\sqrt{13}}{2} & 0 \end{array}\right]$$

$$\Big\downarrow \begin{array}{l} r_1 = \left(\frac{1}{2}+\frac{\sqrt{13}}{2}\right)^{-1} r_1 \\ r_2 = r_2 - \left(\frac{1}{2}+\frac{\sqrt{13}}{2}\right)^{-1} r_1 \\ r_3 = r_3 - 4\left(\frac{1}{2}+\frac{\sqrt{13}}{2}\right)^{-1} r_1 \end{array}$$

$$\left[\begin{array}{ccc|c} 1 & -\frac{1}{2}+\frac{\sqrt{13}}{2} & \frac{1}{6}-\frac{\sqrt{13}}{6} & 0 \\ 0 & 1 & -\frac{11}{18}-\frac{\sqrt{13}}{18} & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \xleftarrow[]{\begin{array}{l} r_2 = -\frac{1}{3} r_2 \\ r_3 = r_3 + \frac{1-2\sqrt{13}}{3} r_2 \end{array}} \left[\begin{array}{ccc|c} 1 & -\frac{1}{2}+\frac{\sqrt{13}}{2} & \frac{1}{6}-\frac{\sqrt{13}}{6} & 0 \\ 0 & -3 & \frac{\sqrt{13}}{6}+\frac{11}{6} & 0 \\ 0 & 1-2\sqrt{13} & \frac{5}{6}+\frac{7\sqrt{13}}{6} & 0 \end{array}\right]$$

$$\Big\downarrow r_1 = r_1 - \left(-\frac{1}{2}-\frac{\sqrt{13}}{2}\right) r_2$$

$$\left[\begin{array}{ccc|c} 1 & 0 & \frac{2}{9}+\frac{\sqrt{13}}{9} & 0 \\ 0 & 1 & -\frac{11}{18}-\frac{\sqrt{13}}{18} & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$$\therefore \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_3 \begin{bmatrix} -\frac{2}{9}-\frac{\sqrt{13}}{9} \\ \frac{11}{18}+\frac{\sqrt{13}}{18} \\ 1 \end{bmatrix} = \frac{x_3}{18} \begin{bmatrix} -4-2\sqrt{13} \\ 11+\sqrt{13} \\ 18 \end{bmatrix}$$

$$\therefore \text{eigenvector } v_3 = \begin{bmatrix} -4-2\sqrt{13} \\ 11+\sqrt{13} \\ 18 \end{bmatrix}$$

3(b) $\text{nul } B = \text{eigenspace of } B \text{ for } \lambda = 0$

$$= \text{span}\left\{ \begin{bmatrix} -4 \\ 5 \\ 7 \end{bmatrix} \right\}$$

$$\text{col } B = \text{span}\left\{ \begin{bmatrix} 2 \\ 1 \\ 4 \end{bmatrix}, \begin{bmatrix} 3 \\ -2 \\ -1 \end{bmatrix} \right\}$$

$v_1 \in \text{nul } B$, $v_2 \in \text{col } B$

$$\Rightarrow v_1 + v_2 \in \text{span}\left\{ \begin{bmatrix} -4 \\ 5 \\ 7 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 4 \end{bmatrix}, \begin{bmatrix} 3 \\ -2 \\ -1 \end{bmatrix} \right\}$$

$$\left[\begin{array}{ccc|c} -4 & 2 & 3 & 0 \\ 5 & 1 & -2 & 0 \\ 7 & 4 & -1 & 0 \end{array}\right] \xrightarrow[r_3 = r_3 + \frac{7}{4} r_1]{\substack{r_1 = -\frac{1}{4} r_1 \\ r_2 = r_2 + \frac{5}{4} r_1}} \left[\begin{array}{ccc|c} 1 & -1/2 & -3/4 & 0 \\ 0 & 7/2 & 7/4 & 0 \\ 0 & 15/2 & 17/4 & 0 \end{array}\right] \xrightarrow[r_3 = 2(r_3 - \frac{15}{2} \cdot \frac{2}{7} r_2)]{r_2 = \frac{2}{7} r_2} \left[\begin{array}{ccc|c} 1 & -1/2 & -3/4 & 0 \\ 0 & 1 & 1/2 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right]$$

$$\xrightarrow{\substack{r_2 = r_2 - \frac{1}{2} r_3 \\ r_1 = r_1 + \frac{3}{4} r_3}} \left[\begin{array}{ccc|c} 1 & -1/2 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right] \xrightarrow{r_1 = r_1 + \frac{1}{2} r_2} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right]$$

$\therefore \begin{bmatrix} -4 \\ 5 \\ 7 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 4 \end{bmatrix}, \begin{bmatrix} 3 \\ -2 \\ -1 \end{bmatrix}$ are linearly independent

$\therefore$ they form a basis in $\mathbb{R}^3$

Hence, every vector in $\mathbb{R}^3$ can be uniquely written as a sum of v_1 and v_2

3(c) let $T(x) = A(x)$ where A is the transformation matrix of T

$T(v_1 + v_2) = v_2$

$\Leftrightarrow T\left(a\begin{bmatrix}-4\\5\\7\end{bmatrix} + b\begin{bmatrix}2\\1\\4\end{bmatrix} + c\begin{bmatrix}3\\-2\\-1\end{bmatrix}\right) = b\begin{bmatrix}2\\1\\4\end{bmatrix} + c\begin{bmatrix}3\\-2\\-1\end{bmatrix}$ where $a, b, c \in \mathbb{R}$

$$T\left(\begin{bmatrix}-4\\5\\7\end{bmatrix}\right) = A\begin{bmatrix}-4\\5\\7\end{bmatrix} = \begin{bmatrix}0\\0\\0\end{bmatrix}$$

$$T\left(\begin{bmatrix}2\\1\\4\end{bmatrix}\right) = A\begin{bmatrix}2\\1\\4\end{bmatrix} = \begin{bmatrix}2\\1\\4\end{bmatrix}$$

$$T\left(\begin{bmatrix}3\\-2\\-1\end{bmatrix}\right) = A\begin{bmatrix}3\\-2\\-1\end{bmatrix} = \begin{bmatrix}3\\-2\\-1\end{bmatrix}$$

$$\Leftrightarrow A\begin{bmatrix}-4 & 2 & 3\\5 & 1 & -2\\7 & 4 & -1\end{bmatrix} = \begin{bmatrix}0 & 2 & 3\\0 & 1 & -2\\0 & 4 & -1\end{bmatrix}$$

$$\therefore A = \begin{bmatrix}0 & 2 & 3\\0 & 1 & -2\\0 & 4 & -1\end{bmatrix}\begin{bmatrix}-4 & 2 & 3\\5 & 1 & -2\\7 & 4 & -1\end{bmatrix}^{-1}$$

$$= \begin{bmatrix}0 & 2 & 3\\0 & 1 & -2\\0 & 4 & -1\end{bmatrix}\begin{bmatrix}-1 & -2 & 1\\9/7 & 17/7 & -1\\-13/7 & -30/7 & 2\end{bmatrix}$$

$$= \begin{bmatrix}-3 & -8 & 4\\5 & 4 & -5\\7 & 14 & -6\end{bmatrix}$$

4(a) let $A = [v_1 \ v_2 \ v_3]$

$$= \begin{bmatrix} 1 & -1 & 2 \\ 1 & 1 & 5 \\ 3 & -1 & 10 \\ 3 & -3 & 6 \end{bmatrix}$$

let $Ax = 0$

$$\left[\begin{array}{ccc|c} 1 & -1 & 2 & 0 \\ 1 & 1 & 5 & 0 \\ 3 & -1 & 10 & 0 \\ 3 & -3 & 6 & 0 \end{array}\right] \xrightarrow[r_4 = r_4 - 3r_1]{\substack{r_2 = r_2 - r_1 \\ r_3 = r_3 - 3r_1}} \left[\begin{array}{ccc|c} 1 & -1 & 2 & 0 \\ 0 & 2 & 3 & 0 \\ 0 & 2 & 4 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \xrightarrow[r_3 = r_3 - r_2]{r_2 = \frac{1}{2} r_2} \left[\begin{array}{ccc|c} 1 & -1 & 2 & 0 \\ 0 & 1 & 3/2 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$$\xrightarrow{\substack{r_2 = r_2 - \frac{3}{2} r_3 \\ r_1 = r_1 - 2r_3}} \left[\begin{array}{ccc|c} 1 & -1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \xrightarrow{r_1 = r_1 + r_2} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$\therefore$ The pivot columns are 1, 2, 3

$\therefore$ $c1, c2, c3$ of A are linearly independent

$\therefore$ v_1, v_2, v_3 are linearly independent

(b)

$$\left[\begin{array}{cccc|c} 1 & -1 & 2 & 1 & 0 \\ 1 & 1 & 5 & 2 & 0 \\ 1 & -1 & 10 & 4 & 0 \\ 3 & -3 & 6 & p & 0 \end{array}\right] \xrightarrow[r_4 = r_4 - 3r_1]{\substack{r_2 = r_2 - r_1 \\ r_3 = r_3 - 3r_1}} \left[\begin{array}{cccc|c} 1 & -1 & 2 & 1 & 0 \\ 0 & 2 & 3 & 1 & 0 \\ 0 & 2 & 4 & 1 & 0 \\ 0 & 0 & 0 & p-3 & 0 \end{array}\right]$$

$$\xrightarrow[r_3 = r_3 - r_2]{r_2 = \frac{1}{2} r_2} \left[\begin{array}{cccc|c} 1 & -1 & 2 & 1 & 0 \\ 0 & 1 & 3/2 & 1/2 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & p-3 & 0 \end{array}\right]$$

For $\{v_1, v_2, v_3, v_4\}$ to be linearly independent, the linear system must contain 4 pivots, ie.

$p - 3 \neq 0$.

$p \neq 3$

$\therefore$ range of $p = \mathbb{R} \setminus \{3\}$

(c) $v_2 \neq k v_3$ where $k \in \mathbb{R}$

$\Rightarrow \{v_2, v_3\}$ are linearly independent

$\therefore$ dim span $\{v_2, v_3\} = 2$

5(a) $v_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ $v_2 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ $S = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ $E = \begin{bmatrix} 8 \\ 10 \end{bmatrix}$

let $E - S = a v_1 + b v_2$

$$\begin{bmatrix} 7 \\ 9 \end{bmatrix} = a \begin{bmatrix} 1 \\ 2 \end{bmatrix} + b \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} \text{ where } a, b \in \mathbb{N}$$

$$\left[\begin{array}{cc|c} 1 & 2 & 7 \\ 2 & 1 & 9 \end{array}\right] \xrightarrow{r_2 = r_2 - 2r_1} \left[\begin{array}{cc|c} 1 & 2 & 7 \\ 0 & -3 & -5 \end{array}\right]$$

$$\downarrow r_2 = -\tfrac{1}{3} r_2$$

$$\left[\begin{array}{cc|c} 1 & 0 & 11/3 \\ 0 & 1 & 5/3 \end{array}\right] \xleftarrow{r_1 = r_1 - 2r_2} \left[\begin{array}{cc|c} 1 & 2 & 7 \\ 0 & 1 & 5/3 \end{array}\right]$$

$$\Rightarrow a = \frac{11}{3}, \quad b = \frac{5}{3}$$

$\therefore$ a, b are not natural numbers, thus it is a contradiction

$\therefore$ no path exists between S and E

(b) $\begin{aligned} x - 2y - z &= 0 \\ 3x - 3y + z &= 0 \end{aligned} \iff \begin{bmatrix} 1 & -2 & -1 \\ 3 & -3 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

$$\left[\begin{array}{ccc|c} 1 & -2 & -1 & 0 \\ 3 & -3 & 1 & 0 \end{array}\right] \xrightarrow{r_2 = r_2 - 3r_1} \left[\begin{array}{ccc|c} 1 & -2 & -1 & 0 \\ 0 & 3 & 4 & 0 \end{array}\right] \xrightarrow{r_2 = \frac{1}{3} r_2} \left[\begin{array}{ccc|c} 1 & -2 & -1 & 0 \\ 0 & 1 & 4/3 & 0 \end{array}\right]$$

$$\downarrow r_1 = r_1 + 2r_2$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 5/3 & 0 \\ 0 & 1 & 4/3 & 0 \end{array}\right]$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -\frac{5}{3} z \\ -\frac{4}{3} z \\ z \end{bmatrix} = \frac{z}{3} \begin{bmatrix} -5 \\ -4 \\ 3 \end{bmatrix}$$

let $r(t)$ be the line of intersection

$\therefore r(t) = t \begin{bmatrix} -5 \\ -4 \\ 3 \end{bmatrix}$ where $t \in \mathbb{R}$